

Guided Notes: The Quadratic Formula

Name: _____

Algebra I | Guided notes — we fill these in together

Vocabulary

Standard Form: A quadratic equation written as _____, where $a \neq 0$.

Quadratic Formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$, a formula that solves any equation written in _____ form.

Discriminant: The expression _____, found under the radical, which tells how many real solutions an equation has.

We do together. Solve $2x^2 + 7x - 4 = 0$ using the quadratic formula.

Step 1. Identify the coefficients: $a =$ _____, $b =$ _____, $c =$ _____.

Step 2. Compute the discriminant: $b^2 - 4ac = 7^2 - 4(2)(-4) = 49 + 32 =$ _____.

Step 3. Take the square root: $\sqrt{81} =$ _____.

Step 4. Substitute into the formula: $x = \frac{-7 \pm 9}{2(2)} = \frac{-7 \pm 9}{4}$.

Step 5. Simplify both solutions: $x =$ _____ or $x =$ _____.

You Try. Solve each. Show every step, then name the answer.

a) Solve $x^2 + 8x + 15 = 0$ using the quadratic formula.

b) Solve $x^2 - 9x + 20 = 0$ using the quadratic formula.

Summary (fill in): To solve a quadratic equation, first write it in standard form $ax^2 + bx + c = 0$, then substitute a , b , and c into $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$. The discriminant $b^2 - 4ac$ predicts the outcome: if it is _____ there are two real solutions, if it is _____ there is exactly one real solution, and if it is _____ there are no real solutions.